Physics-Informed Double Deep Q-Networks for Scalable and Energy-Efficient EV Routing

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***Abstract*— Standard Deep Q-Networks (DQN) depend on trial-and-error exploration. In electric vehicle (EV) routing, this often leads to violations of physical constraints, since battery depletion is only detected after it has occurred. To address this limitation, this paper proposes a Physics-Informed Double Deep Q-Network (PI-DDQN) that evaluates thermodynamic feasibility before any action is chosen. The network operates on realistic urban topologies produced by a Spatial Graph-Generative Adversarial Network (SG-GAN), where a hard physics mask filter discards every routing action that would drain the fleet's batteries. Evaluations across dynamic map complexities and congestion levels demonstrate that the proposed method achieves an optimal energy consumption of 20.70 kWh/100 km. Crucially, it reduces physical constraint violations from 1,845 in tabular baselines to zero and improves the mission success rate from 65.4% to 100%. Training converges rapidly in 800 episodes, maintaining an inference time under 0.2 ms and a highly constrained memory footprint of 350 KB for 100-node networks.**

***Index Terms*—Electric Vehicle Routing, Standard DQN, PI-DDQN, Action Masking, SG-GAN**

I. INTRODUCTION

Nowadays, the environmental impact of industrial supply chains is under increasing scrutiny. According to the sixth assessment report of Intergovernmental Panel on Climate Change (IPCC), the global transportation sector produces 15% of all global emissions, which is roughly 8.9 $GtCO_2$ eq. The last-mile delivery fleets of many delivery service companies using gasoline-based vehicles are large producers of urban air pollution. To solve this, these companies are transiting to electric vehicles (EVs) to meet net-zero commitments [1]. With multiple schemes and policies of the governments to support EV adoption, Large-scale transition modelling proposes that the market share of EVs will increase by 70–85% [2]. Consequently, this means more load on electricity grids to meet the demands of charging stations for EVs, reducing the estimated impact in emission reduction.

At the operational level, EVs introduce constraints that internal combustion vehicles do not face: restricted per-charge range, longer refuelling times, and sparse charging infrastructure. Taken together, these constraints define the Electric Vehicle Routing Problem (EVRP): given a fleet of EVs with limited range, routes decided should minimize total travel distance while ensuring that no vehicle exhausts its battery between charging stops.

Planning routes for electric delivery trucks has become too complex for legacy software. Modern route planners aim to factor in terms like 'payload', 'aerodynamic drag', and 'drivetrain losses' [3], and infrastructure is expanding to include things like battery-swapping and wireless charging lanes. If there is any unexpected change in traffic, then they have to calculate it right from the start again, as there is no real time information.

Rest of the paper is structured as follows: Literature review is provided in Section II. Section III describes the methodology of optimization. The results and computational experiments are then detailed in Sections IV and V . Finally, Section VI concludes the study.

II. LITERATURE REVIEW

Effectiveness of EV routing system greatly depends on planning paths within strict time and battery limits, and predicting step-by-step energy consumption accurately enough to avoid failures in the middle of a journey.

Regular grid maps cannot replicate the node-degree variability, irregular edge lengths, and uneven connectivity found in real cities, and thus reduce the value of benchmarks. On the other hand, generative machine learning methods synthesise graphs whose statistical properties are closer to those of actual urban networks. Das et al. used (SG-GAN) with tabular RL to generate real urban road maps and reduced vehicle energy usage [4].

Metaheuristic methods have also been used for pathfinding under hard operational constraints. Genetic algorithms combined with A* search have addressed the multi-objective EVRP under tight time windows [5]. But changing traffic conditions force these methods to restart their entire pipeline, limiting their suitability for real-time navigation.

Deep Reinforcement Learning (DRL) instead treats routing as a sequential decision problem. Wu et al. showed that DRL controllers adapted speed and trajectory to shifting traffic, consuming less energy than fixed-policy methods [6]. Deep Q-networks (DQN) trained at signalled intersections cut losses from stop-and-go vehicle platoons

[7]. DQN has also supported adaptive cruise control by limiting battery drain during traffic slowdowns [8].

Although DQN performs well at dynamic routing decisions, overestimation bias and a fundamental blindness to physics remain its weaknesses, letting agents choose infeasible routes that go unpenalized until the battery is already depleted. Double DQN (DDQN) architectures address the overestimation issue but still lack physical safety bounds. To bridge this gap, this paper proposes a Physics-Informed Double Deep Q-Network (PI-DDQN) that draws on Safe Reinforcement Learning principles to apply a strict thermodynamic action mask before action selection. Constraint violations are thereby eliminated mathematically, without sacrificing the scalability that massive urban road networks demand.

III. METHODOLOGY

The proposed framework unfolds in three sequential stages. A road network is first synthesised using the SG-GAN, after which the physical energy and dynamic congestion models governing this environment are established these in turn generate the per-step cost signals. The PI-DDQN is then trained inside the resulting physics-bound environment, with thermodynamic constraints folded directly into action selection.

A. *Environment Synthesis via SG-GAN*

Within the proposed framework, SG-GAN [9] synthesises urban street structures directly from empirical OpenStreetMap (OSM) spatial data. The topologies that emerge are nonplanar and highly realistic, and they serve as the simulation environment in which the PI-DDQN model is trained, as summarised in Table I.

TABLE I: SIMULATION PARAMETER REFERENCE

Parameter	Value
Map Location	28.6193◦N, 77.0334◦E
Map Radius	800 m
Default Junctions / CS	35, 43, 50 Nodes / 7, 9, 10 CS

B. *Physics-Informed Double Deep Q-Network*

Once the SG-GAN has synthesised the road topology, the PI-DDQN is deployed on it. Three mechanisms are combined in this single architecture: neural function approximation, which keeps the method tractable as the graph scales; Double DQN target decoupling [10], which removes the overestimation bias driving unconstrained agents toward costly routes; and an energy-based physical mask that blocks infeasible actions before selection. PI-DDQN hyperparameters are listed in Table IV.

1) System Architecture and State Representation: At each decision step ***t***, the state for an EV on graph G (V, E) is given by equation 1:

$$s_t = \left[onehot(v_t), onehot(v_{dest}), \frac{(SOC_t)}{(SOC_{max})} \right], \qquad (1)$$

$$dim(s_t) = 2|V| + 1$$

Where *onehot* denotes binary array encoding, $v_t \in V$ is the current node, $v_{dest} \in V$ is the destination, SOC_t is the current battery level, and $SOC_{\max}$ is absolute battery capacity.

For a 50-node topology ($|V| = 50$), the resulting 101-dimensional vector is processed by a 101 → 128 → 128 → 50 feedforward network with rectified linear unit (ReLU) activations [11].

To mitigate MARL non-stationarity, all 50 EVs share a centralized architecture organised into three operational zones (Figure 1). In **Zone 1 (Inference):** The Online Q-Network (θ) computes routing Q-values, and a Physics Mask blocks any battery-draining action before execution; EVs at critically low charge are instead rerouted through Bipartite Allocation [12]. **Zone 2 (Training):** draws mini batches from a Centralized Replay Memory (D) [13] and, using an Adam optimizer, minimize a joint loss ($\mathcal{L}_{\text{total}}$), that weighs route efficiency ($\mathcal{L}_{\text{TD}}$) against physical safety $\left(\mathcal{L}_{\text{phys}}\right)$ .In **Zone 3 (Target Estimation):** a decoupled Target Network ($\theta^- \leftarrow \theta$) stabilizes value estimates, removing the overestimation bias inherit to traditional Q-learning.

2) Physics-Informed Action Masking and Optimization:

At each step, the framework computes energy demand $E_{\text{req}}(a)$ for each adjacent edge and constructs a binary action mask from it, as expressed by equation 2.

$$M(s_t, a) = \begin{cases} 0, & \text{if } \mathrm{SOC_t} \geq \mathrm{E_{req}(a)}, \\ -\infty, & \text{otherwise.} \end{cases} \qquad (2)$$

Masked Q-values are computed as $\tilde{Q}(s_t, a) = Q(s_t, a; \theta) + M(s_t, a)$,which reduces the selection probability of infeasible actions to zero by construction. Training then relies on a masked ε-greedy policy whose random exploration is confined to the thermodynamically feasible action set, so even under aggressive high-ε exploration the agent cannot execute battery-draining paths. The physics penalty $\left(\mathcal{L}_{\text{phys}}\right)$ can therefore shapes the network's energy boundary representation safely, without risking real-world constraint violations [14]. The process ultimately minimizes a joint loss ($\mathcal{L}_{\text{total}}$) merging route efficiency ($\mathcal{L}_{\text{TD}}$) and the physics penalty, as defined in Equation 3:"

$$\mathcal{L}_{phys} = \mathcal{L}_{TD} + \lambda E \left[max\big(0, Q(s_t, a_t; \theta) - \left[p_t + \gamma(1 - d_t)\hat{Q}_{next}\right]\big)^2 \right] \qquad (3)$$

where $p_t = r_{max} - \alpha \left(\frac{E_{req}(a_t)}{E_{norm}}\right)$ $(with\ r_{max} = 10,\ \alpha = 5)$

is the per-transition feasibility bound, and λ weights the physics penalty. This embeds the energy constraint directly into training, so the network no longer needs to learn penalizing infeasible decisions on its own.

Shaped Reward Function: Terminal rewards being sparse, the agent relies on a dense shaped reward instead, given in equation 4.

$$r_t = 10 - 5 \left[\beta \frac{E_{ij}}{E_{\text{norm}}} + (1 - \beta) \frac{t_{ij}}{T_{\text{norm}}} \right] - 2\delta_{ij}$$

$$+1_{\text{dest}} \cdot 100 - 1_{\text{depleted}} \cdot 50 + \Delta\Phi(s_t) \quad (4)$$

The shaped reward draws on normalized edge energy (E_{ij}/E_{norm}) and travel time (t_{ij}/T_{norm}), with $\beta = 0.8$ tilting the balance toward energy conservation while δ_{ij} penalizes local congestion. Binary indicators 1_{dest} and $1_{depleted}$ further assign +100 for reaching the destination and -50 for fatal battery depletion. Policy optimality remains preserved [15] via a potential $\Phi(s)$ based on A* shortest-path heuristics. Its differential, $\Delta\Phi(s_t)$, is scaled $by\ \omega = 0.1$ so that this geometric pull stays a secondary tiebreaker, never overriding the primary thermodynamic objective.

Algorithm 1 gives the full routing process, and Figure 1 shows the agent architecture

Algorithm 1 Multi-Agent PI-DDQN Routing with Physics Masking

Require: Road graph *G* (*V*, *E*), EV count *N*, replay capacity *B*, batch size *b*

```
Initialize online Q(θ), target Q̂(θ⁻) ← Q(θ), replay buffer D
for epoch = 1 to I_π do
    Sample N agents with random (v_start, v_dest, SOC_0), initialize active set A
    for step =1 to T_max do
        for each agent k ∈ A do
            if SOC_k ≤ T_h then
                Reassign destination to the optimal minimum cost charging station
            end if
            Compute energy demand E_req(a) and apply mask M(s_t, a)
            Select action a_t using masked ϵ-greedy policy on Q̃(s_t, a)
            Execute transition: update congestion δ_ij, energy E_ij, and SOC_K
            Store transition (s_t, a_t, r_t, s_{t+1}, d_t, p_t) in D
        end for
        if |D| ≥ b then
            Sample mini batch from D and compute DDQN targets Y_t^DDQN
            Minimize composite loss L_TD + λL_phys via Adam optimizer
        end if
    end for
    if epoch mod τ_sync = 0 then
        Synchronize target network: θ⁻ ← θ
    end if
end for
```

Calculation of Energy Consumption, Travel Time and Congestion

The PI-DDQN evaluates environmental fitness through traction force, dynamic congestion, and charging dwell times. The physical constants used in the energy model appear in Table III

***1) Energy Consumption and Regenerative Braking**:* Edge energy derives from aerodynamic drag, rolling resistance, and road grade as given in equation 5 where the subscript ij denotes the road segment from the current node i to the next node j.

$$F_{ij} = \frac{1}{2}\rho C_d A v_i^2 + C_r mg\cos(\theta_{ij}) + mg\sin(\theta_{ij}) \quad (5)$$

Mechanical power $(P_{ij} = F_{ij} \cdot v_{ij})$ is converted to electrical energy at drivetrain efficiency (η). Net energy $(E_{net,ij})$ integrates edge distance (d_{ij}) and captures regenerative braking, successfully recovering 80% of kinetic energy when decelerating $(F_{ij} < 0)$,as represented by equation 6:

$$E_{net,ij} = \begin{cases} \frac{F_{ij} \cdot d_{ij}}{\eta \cdot (3.6\times10^6)}, & if\ F_{ij} \geq 0 \\ 0.80 \cdot \frac{F_{ij} \cdot d_{ij}}{3.6\times10^6}, & if\ F_{ij} < 0 \end{cases} \quad (6)$$

***2) Dynamic Congestion and Travel Time**:* Using the BPR function [16], the congestion factor (δ_{ij}) evaluates current traffic flow (q_{ij}) against maximum edge capacity (c_{ij})as shown below in equation 7:

$$\delta_{ij} = \min\left(1.0, \frac{q_{ij}}{c_{ij}}\right) \quad (7)$$

Final energy scales with congestion as equation 9, applying up to a 20% penalty at full gridlock and updating SOC_{t+1}. Traversal time (t_{ij}) incorporates congestion delays and charging dwell as represent in equation 8:

$$t_{ij} = t_{ij}^{\text{base}}(1 + 0.5\delta_{ij}) + \frac{B_{\max} - B_{\text{current}}}{\eta_s} \quad (8)$$

Here, t_{ij}^{base} is the free-flow time, while the latter term calculates charging duration using required battery replenishment $(B_{\max} - B_{current})$ and the station's charging rate (η_s).Queueing delays follow Little's Theorem $(W = L/\lambda_{rate})$,dynamically increasing δ_{ij} on edges approaching saturated hubs [17]. Training hyperparameters are listed in Table II.

TABLE II: TRAINING HYPERPARAMETERS

Parameter	Value	Description
Replay Memory (B)	50,000	Maximum capacity of stored transitions
Learning Rate	0.0005	Adam Optimizer rate for weight updates
Batch Size	64	Samples pulled per training step
Physics Lambda (λ)	0.25	Weight of physics regularization in total loss

Together, these sub-models provide a physics-grounded cost at each step, letting PI-DDQN balance energy, time, and charging load within a single reward signal.

IV. RESULTS AND DISCUSSION

All experiments used a mesh road network generated by the SG-GAN, comprising 35, 43, and 50 junction nodes

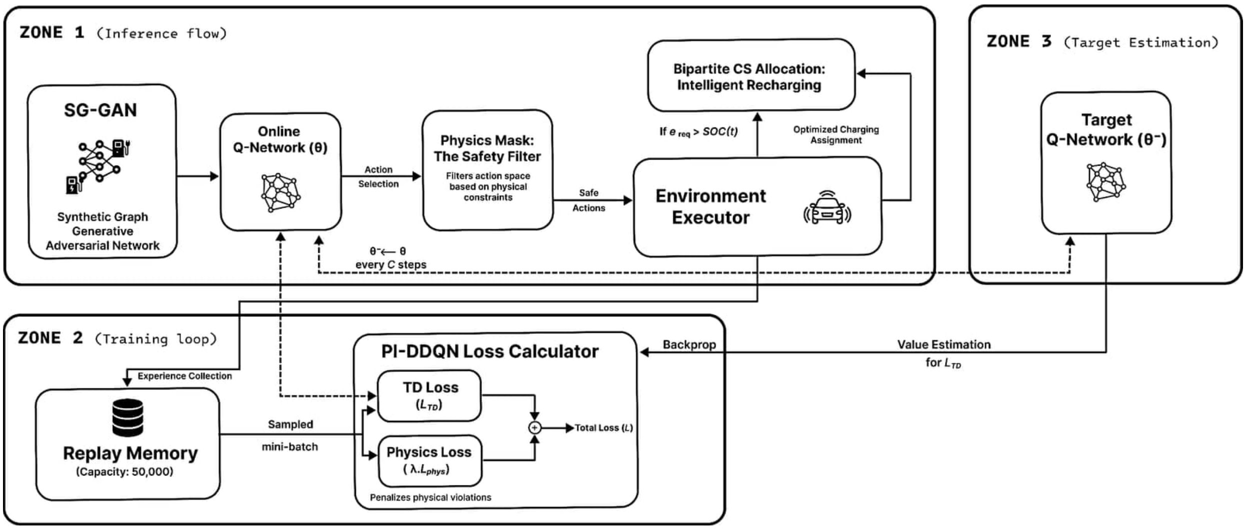

Fig. 1: Physics-Informed Double Deep Q-Network Architecture for Autonomous EV Routing

TABLE III: PARAMETERS RELATED TO ENERGY CALCULATION

Parameter Symbol	Description	Value
ρ	Air density	1.225 kg/m3
Cd	Drag coefficient	0.28
A	Frontal area of the vehicle	2.3 m2
Cr	Rolling resistance coefficient	0.01
m	Mass of the EV	1500 kg
g	Acceleration due to gravity	9.81 m/s2
η	Mechanical to electrical efficiency	0.85
θ	Angle of the incline	0°–3° (random)
I'	Congestion penalty multiplier	+20% at I' = 1.0

(with 7, 9, and 10 charging stations, respectively), and multiple number of EVs which is discussed in Table **V.** Simulations ran in a Python-based Jupyter environment on an HP Victus 16 laptop equipped with an NVIDIA GeForce RTX 3050 (6 GB) GPU and 16 GB of RAM.

Varying the node count in SG-GAN produces three distinct road networks (Fig. 2): a 35-node layout, a 43-node intermediate grid, and a 50-node dense urban structure. Each EV starts with a 10-kWh battery, with initial SOC drawn uniformly from [30%, 100%], giving a mean of 65% and a standard deviation of 20.2%.

Whenever an EV's SOC drops below 25%, it is rerouted to the nearest charging station, which corresponds to a 2.5 kWh reserve on the 10-kWh battery. For larger batteries, the limit drops to 16.7% for a 15-kWh battery, 12.5% for 20 kWh, and 8.3% for 30 kWh. This ensures the emergency reserve remains the same, which prevents the vehicles from taking unnecessary charging trips.

A. Network Topology Validation

To ensure any performance shifts were affected only by topology scale, all three graph setups operated under identical physical parameters. The 50-node network increases the number of valid actions available at each node. The PI-DDQN performance not dropping off shows that the policy adapts to different connectivities without experiencing state-space degradation.

TABLE IV: PI-DDQN Model Configuration Hyperparameters

Parameter	Description	Value
Network Type	Network used for experiments	Mesh
Initial Nodes	Initial number of nodes in the network	35, 43, 50
Maximum Episode	Cutoff training episode number	800
Learning Rate (α)	Q-value update step size (Adam)	0.0005
Exploration	Initial exploration rate in ϵ-greedy	1.0
Cutoff Exploration	Cutoff exploration rate in ϵ-greedy	0.05
Gamma (γ)	Discount factor	0.95
Beta (β)	Weighting factor	0.8
Batch size	Replay buffer sample size	64
Physics (λ)	Penalty regularization weight	0.25
Replay Buffer (D)	Experience replay capacity	50,000
T_h	EV battery SOC threshold	25%
B_{Max}	EV battery capacity	10 kWh

B. Experimental Setup and Dynamic Congestion Modelling

Each algorithm was evaluated on the same artificial network with the same physical parameters. Congestion is not a simple speed penalty. It changes based on two factors, global vehicle activity and local edge saturation.

Step 1 – Congestion Level (Global): The simulation tracks total EVs traveling across the network at any given time to capture macroscopic traffic conditions. The discrete traffic states are defined in Table V

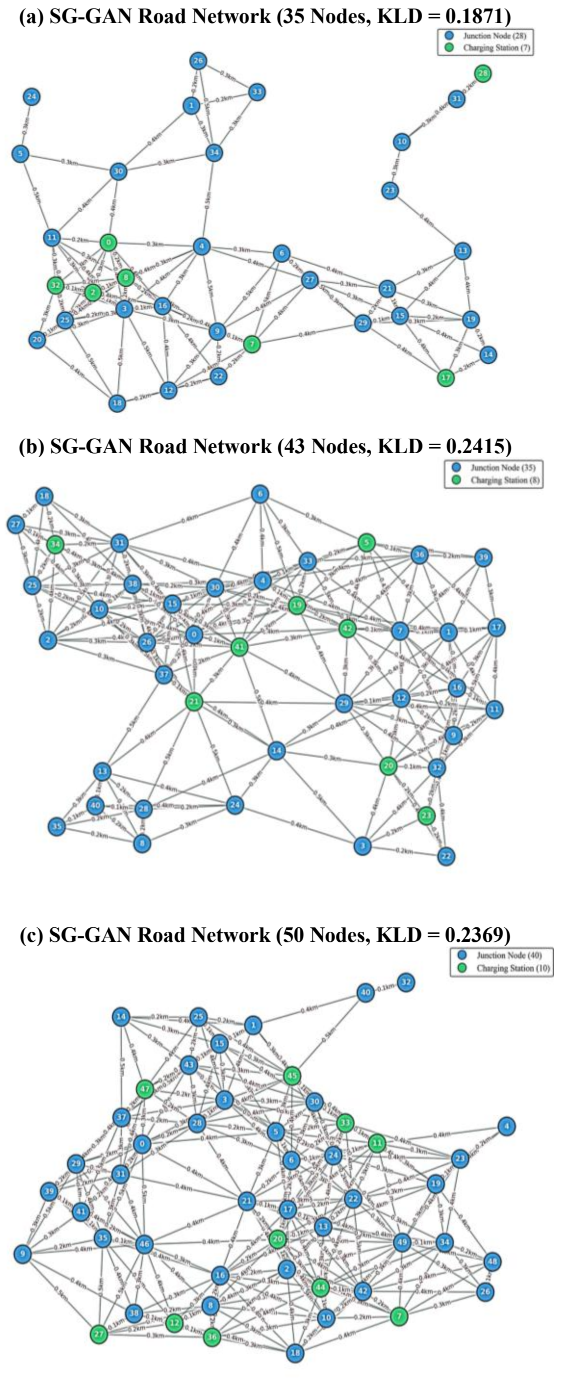

Fig. 2: Three SG-GAN generated road networks illustrating varying topological complexities.

TABLE V: DYNAMIC CONGESTION LEVEL SCALING

Congestion Level	Active EVs	Formula
25% (Light Traffic)	12 EVs	max (1, int (50 × 0.25))
50% (Heavy Traffic)	25 EVs	max (1, int (50 × 0.50))
100% (Gridlock)	50 EVs	max (1, int (50 × 1.00))

Step 2 – Per-Edge Local Congestion Factor (δ_{ij}): Each EV transition increments a shared congestion counter on the traversed edge. The local stop-and-go congestion factor is continuously evaluated as $\delta_{ij} = \min(1.0, \text{flow}_{ij}/\text{capacity}_{ij})$. This factor directly inflates energy consumption at the physics layer as given in equation 9:

$$E_{final,ij} = E_{base,ij} \times (1.0 + 0.2 \times \delta_{ij}) \qquad (9)$$

At full gridlock (δ = 1.0), energy consumption is 20% higher than on a free-flowing network, and the congestion counter decays each timestep to reflect natural traffic dissipation. Drivetrain efficiency (0.85) and drag coefficient (0.28) are fixed to measured physical values, which stops the model from converging onto routes that would be infeasible outside the simulation.

C. Algorithm Superiority and Theoretical Optimality

Standard A^* and the other traditional methods in Table VI do not account for live traffic conditions, which regularly pushes vehicles onto high-energy routes. Learning-based baselines such as DQN and GNN-RL run into overestimation problems and have no mechanism to enforce physical constraints, pulling their results away from the optimum. PI-DDQN reaches 20.70 kWh/100 km by combining double Q-network target decoupling with physics-informed action masks. Testing all baselines under the same conditions proved that PI-DDQN was the most energy-efficient method across every configuration.

D. Ablation Study and Physical Safety Guarantees

Without physical constraints, tabular Q-learning yields 1,845 violation cases. This shows that the route needed more energy than the remaining SOC of the vehicle. The unconstrained DDQN solved the issue of overestimation bias but still had 820 violations. The double-network structure alone does not address the physics-blindness problem.

E. Hyperparameter Optimization

A sensitivity sweep analysed the physics regularization weight λ to confirm the chosen value. Table VII demonstrates that setting $\lambda = 0.25$ uniquely eliminates all violations while keeping energy consumption minimized at 20.70 kWh/100 km.

Table VIII compares energy consumption across three traffic load levels for each topology. In the 50-node network scenario, the Q-learning algorithm shows a steady consumption increase from 22.56 to 23.57 kWh/100 km as congestion worsens. This trend highlights the algorithm's inability to foresee traffic accumulation before locking into a route. The PI-DDQN displays a different pattern with maximum consumption at a 50% load before dropping again during complete gridlock. At maximum congestion, the action mask completely removes heavily saturated roads from the available choices. As a result, the agent is forced onto secondary paths which are more energy-efficient, despite being physically longer.

F. Training Convergence and Stability

Fig. 3 plots the training convergence of the evaluated models. The tabular baseline shows significant reward variance over 10,000 episodes due to its lack of guided exploration. On the other hand, the PI-DDQN model stabilizes around an episodic reward of −25 within roughly 800 episodes.

TABLE VI: PERFORMANCE OF EV ROUTING ALGORITHMS IN DYNAMIC CONGESTION (50-NODE TOPOLOGY)

Method	Energy Cons. (kWh/100 km)	Training / Inference Time	Key Features	Limitations
Dijkstra's [18]	34.44	Single-pass (< 1 ms)	Guarantees shortest distance	Ignores traffic; high energy drain
Energy A^* [19]	23.80	Single-pass (< 5 ms)	Hand-crafted heuristics	Static queues cause 45 violations
Metaheuristic [20]	26.74	~30 min (100 iters)	Multi-objective balance	Often traps in local optima
Tabular Q-Learn [21]	22.56	~2.5 h (10,000 eps)	Model-free stochastic adaptation	Memory explosion (~78 MB)
Standard DQN [22]	21.80	~4.5 h	Continuous neural mapping	Overestimation; chooses invalid routes
Standard DDQN [23]	21.10	~4.0 h	Double-Q target correction	Physics-blind; 820 battery violations
GNN-RL (SOTA) [24]	24.75	~6.0 h / < 10 ms	Native spatial dependencies	Complex; lacks physics-bounding
Proposed PI-DDQN	**20.70**	**~1.5 h / < 0.2 ms**	**Physics-informed action masks**	**Requires accurate physical variables**

This value means a successful arrival. As defined in Eq. 4, an optimal trajectory accrues roughly −125 in stepwise penalties for distance, energy consumption, and congestion. This penalty is offset by the +100 terminal arrival reward, resulting in a net score of −25. The PI-DDQN achieves this rapid convergence primarily because the action mask eliminates impossible transitions, stopping the agent from wasting time on invalid, battery-draining paths during the early stages of training.

To ensure reliability, five distinct random seeds were used to average the reported metrics. The converged reward reached −25, with a 95% confidence interval of ±0.42, indicating the trained policy is stable.

TABLE VII: SENSITIVITY ANALYSIS OF λ (50 NODE)

Physics Weight (λ)	Energy	Violations
$\lambda = 0.0$ (Unconstrained)	21.10 kWh	820
$\lambda = 0.1$ (Under-penalized)	20.95 kWh	145
$\lambda = 0.25$ **(Optimal)**	**20.70 kWh**	**0**
$\lambda = 0.5$ (Over-penalized)	21.90 kWh	0

TABLE VIII: CONGESTION ADAPTATION ACROSS SCALED NETWORKS

Map	Congestion	Q-Learn	PI-DDQN	Gain
35N	25% (Light)	22.17	20.34	**+8.25%**
	50% (Heavy)	22.34	20.97	**+6.13%**
	100% (Gridlock)	23.17	20.35	**+12.17%**
43N	25% (Light)	22.65	20.78	**+8.26%**
	50% (Heavy)	22.82	21.42	**+6.13%**
	100% (Gridlock)	23.67	20.79	**+12.18%**
50N	25% (Light)	22.56	20.70	**+8.24%**
	50% (Heavy)	22.73	21.34	**+6.11%**
	100% (Gridlock)	23.57	20.71	**+12.13%**

V. COMPUTATIONAL COMPLEXITY AND SCALABILITY

A. *Time and Space Complexity*

For calculating computational efficiency, |V| is total junction nodes, I training epochs, n active EV agent, agents, T maximum episode length, H hidden layer size, B batch size, and d_s state vector dimension. Equations 10 & 11 define the theoretical bounds for time and space:

Time Complexity:

$$O(I \cdot N \cdot T \cdot |V| \cdot H + H^2 + |V| \, log \, |V|) \qquad (10)$$

Space Complexity:

$$O(|V| \cdot H + H^2 + B \cdot d_s) \qquad (11)$$

Tabular models must evaluate every possible state-action pair, whereas PI-DDQN relies on neural approximation, keeping operational costs polynomial. Matrix transformations tied to the network architecture govern state processing, keeping the algorithm fast enough for real-time onboard use.

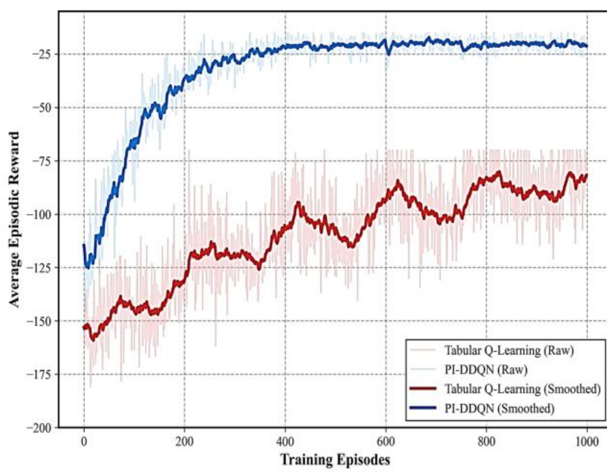

Fig. 3: Training convergence comparison, showing PI-DDQN's path relative to tabular Q-learning.

B. *Memory Footprint Analysis*

Tabular approaches suffer from the curse of dimensionality, forcing memory requirements to grow exponentially, O (|S| × |A|) as the map scales.

Fig. 4 shows a marked gap in resource consumption between the two approaches. On a 35-node network, the baseline Tabular Q-Learning requires ≈ 3.5×10^3 KB. Expanding to 100 nodes causes the memory footprint to skyrocket to nearly 10^5 KB (100 MB). This makes tabular methods impractical for onboard vehicle hardware.

The PI-DDQN framework scales linearly by generalizing states through a fixed-size neural network. This keeps the memory footprint incredibly tight, using only about 215 KB at 35 nodes and maxing out around 350 KB for a 100-node layout. This keeps memory within safe limits and eliminate any danger of memory overflow when deployed on actual hardware.

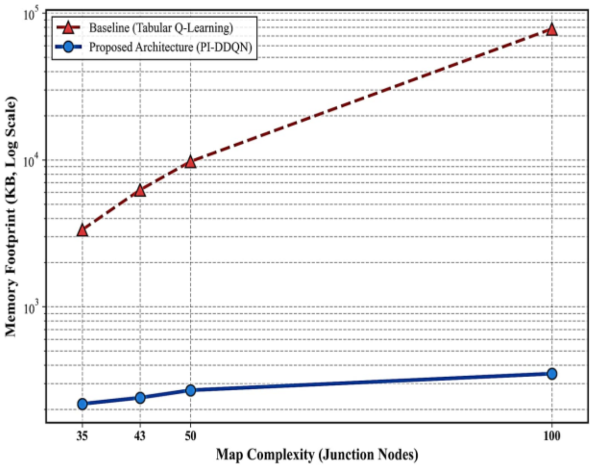

Fig. 4: Memory footprint scaling comparison across varying map complexities.

VI. CONCLUSION

Traditional reinforcement learning for routing relies on penalty mechanisms, meaning the agent learns through repeated vehicle failure. The proposed PI-DDQN framework resolves this by computing edge-specific energy costs and blocking actions that exceed the vehicle's available state of charge prior to selection. This physics-informed masking completely eliminated constraint violations, which dropped from 1,845 under the tabular Q-learning baseline to zero, and improved the mission success rate from 65.4% to 100%.

A fixed-size neural architecture brings the memory footprint for a 100-node network down from roughly 100 MB to 350 KB, with inference times under 0.2 ms. The SG-GAN generator recorded a Kullback-Leibler Divergence of 0.18 on real-world topological data, which supports the validity of the simulation environment. Future work will look at incorporating live traffic feeds and testing the framework on mixed-capacity commercial fleets in previously unseen city maps. Taken together, the results show that embedding physical constraints into the learning process can bring constraint violations to zero while keeping the system practical for real-time EV routing.

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